

Thien (Theo) Pham

400 N Negley Ave Apt 111, Pittsburgh, PA 15206 | +1.919.448.5730 | quythien14@gmail.com
linkedin.com/in/quythien | Google Scholar

EDUCATION

University of Pittsburgh

August 2021 – October 2026

Ph.D. Candidate in Biostatistics (GPA: 3.85)

University of North Carolina at Chapel Hill

September 2017 – May 2021

Bachelor of Science in Statistics, Bachelor of Science in Mathematics (Highest Distinction, GPA: 3.85)

Honors: Phi Beta Kappa, Dean's List

SKILLS

Programming: R (package development, Shiny), Python (PyTorch, PyTorch Geometric, Scanpy), SAS, SQL, STATA, Bash, high-performance and cloud computing (AWS); familiar with Java and C++.

Statistical Methods: Bayesian hierarchical modeling (MCMC), circular statistics, FDR control, causal inference and mediation, longitudinal and survival modeling, missing data methods, simulation-based study design, machine learning.

Omics: bulk and single-cell RNA-seq, proteomics, DNA methylation, multi-omics integration.

Drug Development: *Early development:* companion diagnostics, reader reproducibility, CTA-CDx bridging for time-to-event endpoints. *Late development:* Phase 2–3 surrogate endpoints, Bayesian meta-analysis, MNAR sensitivity analyses.

PROFESSIONAL EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA

Starting November 2026

Special Faculty Postdoctoral Researcher, Department of Statistics & Data Science (Advisor: Kathryn Roeder)

- Research on statistical and machine learning methods for scientific problems in genetics, using large-scale genomics data including single-cell sequencing, proteomics, and metabolomics.

Merck & Co., Inc., Rahway, NJ

June 2026 – August 2026

Early Development Statistician Intern, Biostatistics and Research Decision Sciences (BARDS)

- Supported the Companion Diagnostics and Translational Oncology team through statistical research and method development for biomarker-driven applications in oncology drug development.
- Conducted statistical research on reader reproducibility (RR) in companion diagnostic (CDx) studies supporting Premarket Approval, developing model-based approaches to assess inter-pathologist variability.
- Developed a Bayesian hierarchical framework with partial pooling to stabilize agreement estimates and guide reproducibility study design and sample size planning; implemented it in an internal R package.
- Developed corrected CTA-CDx bridging equations for time-to-event endpoints, addressing limitations of the log-hazard-ratio approach and defining estimands for the CDx-positive intended-use population.

Otsuka Pharmaceutical Companies (U.S.), Princeton, NJ

May 2025 – August 2025

Late Development Statistician Intern

- Developed an internal R and SAS package for eGFR-based clinical trial analyses in IgA Nephropathy (IgAN) to evaluate treatment effects on renal progression and support sibeprenlimab development.
- Applied semi-parametric joint modeling and Bayesian meta-analytic surrogacy methods to longitudinal and time-to-event endpoints, strengthening treatment-effect assessment and surrogate endpoint evaluation in Phase 2–3 renal studies.
- Conducted simulation studies to validate surrogate endpoint frameworks for eGFR- and urinary protein-based analyses, supporting late-stage clinical development and regulatory decision-making in the sibeprenlimab program.
- Evaluated missing data mechanisms in a PTSD clinical program for methylone (TSND-201), including MNAR assumptions, and performed sensitivity and tipping-point analyses to assess robustness of study conclusions.

University of Pittsburgh Medical Center (UPMC), Pittsburgh, PA

May 2023 – May 2026

Graduate Research Assistant

Method Development

- Developed BayesRC, a Bayesian framework for comparative circadian analysis that jointly infers gene-level rhythmicity and phase, quantifies cross-condition concordance, and classifies rhythmic gain, loss, and conservation under Bayesian false discovery rate control.
- Developed SCP, a simulation-based sample size planning framework for circadian omics studies under FDR control, released as an R/Shiny package with 160 curated pilot datasets spanning four species and 100+ tissues.

Translational & Computational Biology

- Applied hierarchical Bayesian models, machine learning, and high-throughput computing to analyze high-dimensional multi-omics data and develop concordance metrics for cross-condition and cross-species analysis.
- Streamlined preprocessing workflows for diverse omics datasets, including bulk and single-cell RNA-seq, improving alignment rates by 20% and supporting downstream integrative analyses.
- Integrated multimodal single-cell and cross-omics data using Scanpy, MultiVI, and deep learning frameworks to improve cellular phenotyping and cross-platform interpretation.
- Applied integrative multi-omics approaches to combine DNA methylation, RNA-seq, and proteomics data, deriving methylation–gene–protein modules for biological interpretation.
- Analyzed circadian biomarkers across multiple brain regions, cancer tissues, and model systems to assess sex- and condition-specific effects in humans and mice.
- Designed graph-based machine learning models in PyTorch Geometric to infer gene–disease relationships from biological network data.

University of Pittsburgh, Department of Infectious Disease

August 2023 – Present

Graduate Research Assistant

- Applied mixed-effects models and Generalized Estimating Equations (GEE) to analyze clinical trial data, deriving insights to improve HIV treatment adherence and quality of life, particularly among women workers in industrial zones.
- Implemented longitudinal mediation models to estimate short-term causal effects of dry needling, identifying pain reduction as a significant mediator of disability outcomes.

Hill-Rom, Cary, NC

June 2020 – August 2020

Research Assistant Intern

- Conducted comprehensive exploratory data analysis and literature review to provide the team with key insights for development of machine learning models for predicting pediatric sepsis in ICU settings.

Carolina Center for Neurostimulation, Chapel Hill, NC

August 2019 – December 2019

Undergraduate Research Assistant

- Investigated cortical synchrony between the thalamus and cortex using Python and MATLAB to enhance models visualizing brain interactions, advancing brain stimulation research and understanding of cortical dynamics.

PUBLICATIONS

Statistical Methodology

- [1] **Pham TQ**, Wang H, Xue X, Zhang X, Yu G, Zhang L, Tseng GC. BayesRC: a comparative Bayesian multilevel framework for evaluating circadian synchrony across conditions. [Submitted to *Briefings in Bioinformatics*]
- [2] **Pham TQ**, Wang H, Seney ML, McClung CA, Yu G, Tseng GC. Comprehensive power evaluation and study design for circadian biomarker detection and differential rhythmicity analysis. [Submitted to *Bioinformatics*] R package: github.com/quythien/SCP

Multi-Omics and Translational Psychiatry

- [3] Barko KN, Shelton MA, Kropp DR, **Pham TQ**, Rainville JR, Xue X, Tseng GC, et al. Impact of sex chromosomes and gonad type in stress susceptibility in corticostriatal brain regions. *Proc Natl Acad Sci U S A*. 2026;123(22):e2531920123.
- [4] Almeida Rojo AL, Barnhardt TR, **Pham TQ**, Heim B, Cai L, Tseng GC, Huang YH. Sleep deprivation engages the orexin/hypocretin system to regulate food reward seeking. *Int J Neuropsychopharmacol*. 2025;28(8):pyaf047. doi:10.1093/ijnp/pyaf047
- [5] Snowden A, **Pham T**, Ketchesin K, Seney M. Divergent effects of smoking in striatal subregions in psychosis. *Biol Psychiatry*. 2026;99(10):S375.
- [6] Gober I, Barton D, Awan N, Fan E, Kumar R, Russell A, **Pham TQ**, et al. Exploring temporal dynamics with aromatization after moderate-to-severe TBI and associations with outcome. *J Neurotrauma*. 2023;40(15-16):A69.
- [7] Gober I, Russell A, Shick T, Vagni V, **Pham TQ**, Carlson J, Kochanek P, Wagner A. Systemic administration of soluble glycoprotein 130 promotes cognitive recovery and decreases proinflammatory chemokine inflammation after controlled cortical impact in mice. *J Neurotrauma*. 2023;40(15-16):A68.
- [8] Barko K, **Pham TQ**, Seney M, et al. Sex differences detected in the proteome of MDD subjects. [Under Review]
- [9] Barko K, **Pham TQ**, Seney M, et al. Microglial transcriptional responses to SCVS: brain region- and sex-specific profiles. [In Preparation]

Clinical Epidemiology and Public Health

[10] Murillo C, Cerezo-Téllez E, Torres-Lacomba M, **Pham TQ**, Lluch E, Falla D, Vo TT. Unraveling the mechanisms behind the short-term effects of dry needling: new insights from a mediation analysis with repeatedly measured mediators and outcomes. *Arch Phys Med Rehabil*. 2024;105(12):2269-2276. doi:10.1016/j.apmr.2024.07.016

[11] Ha T, **Pham TQ**, Misra S, Su TWH, Ha L, Cunningham SD, Schensul SL. Impact of multilevel interventions on reducing reasons for drinking among men living with HIV. *AIDS Care*. 2025;37(10):1696-1705. doi:10.1080/09540121.2025.2547030

[12] Ha T, **Pham TQ**, Nguyen N, Shrestha R, Schensul SL. Prevalence and determinants of induced abortion among women migrant workers in an industrial zone in Vietnam: a cross-sectional study. *Medicine (Baltimore)*. 2025;104(40):e45005. doi:10.1097/MD.00000000000045005

[13] **Pham TQ**, Wu Q, Pan C, Ha L, Bui TC, Sujan MSH, Schensul SL, et al. Enhancing quality of life through tailored interventions for men living with HIV. *Qual Life Res*. 2026;35:27.

[14] Ha T, **Pham T**, Sign R, Schensul S. Correlates of depression among men with HIV in India: secondary analysis of endline data from a randomized controlled trial. *Ann Behav Med*. 2026;60:S544.

[15] Ha T, Wu Q, **Pham T**, Schensul S. The role of tailored interventions in improving quality of life among men living with HIV. APHA 2025 Annual Meeting and Expo; 2025.

[16] Ha T, **Pham T**, Pham B, Schensul S. Determinants of psychosocial well-being in female migrant workers. APHA 2025 Annual Meeting and Expo; 2025.

AWARDS

Travel Funding Award, STATGEN Conference	2025
Biostatistics Dean’s Day Doctoral Award	2025
Outstanding Teaching Assistant (TA) Award	2025
Phi Beta Kappa Induction, top 1% , University of North Carolina at Chapel Hill	2021
International Championship of Collegiate A Cappella, Semi-finalist, rank 2nd	2019, 2020
Outstanding Student Scholarship, top 1% , Pham Ngoc Thach University of Medicine	2015, 2016
Excellent Student at City Level in Biology, rank 2nd	2011, 2013, 2014

TEACHING EXPERIENCE

University of Pittsburgh	Years Taught
<i>Department of Biostatistics, Graduate Teaching Assistant</i>	
• BIOST 2086: Mixed Models	Spring 2024
• BIOST 2050: Longitudinal and Clustered Data Analysis	Fall 2023
• BIOST 2041: Introduction to Statistical Methods	Spring 2022
• BIOST 2038: Foundation of Statistical Theory	Fall 2021, Fall 2022
<i>Institute for Clinical Research Education, Graduate Teaching Assistant</i>	
• CLRES 2020: Biostatistics	Summer 2022, Summer 2024
• CLRES 2005: Computational Methods for Clinical Research	Summer 2022, Summer 2024
University of North Carolina at Chapel Hill	
<i>Department of Mathematics, Undergraduate Teaching Assistant</i>	
• MATH 547: Linear Algebra	Fall 2019

MEMBERSHIP & LEADERSHIP

American Statistical Association (Pittsburgh Chapter), <i>Member</i>	2021 – Present
Pittsburgh Men’s Glee Club, <i>Member</i>	2023 – Present
Tar Heel Voices, UNC’s co-ed a cappella group, <i>Fundraising Committee Member</i>	2019 – 2021
Pham Ngoc Thach University of Medicine, School of Medicine, <i>Class President</i>	2014 – 2016